Machine Learning

Basic Concepts

CENEO.pl

question → → answer

program

algorithm

Q-learning: Learn function $Q : \mathcal{X} \times \mathcal{A} \to \mathbb{R}$

Require:

Sates $\mathcal{X} = \{1, \ldots, n_x\}$

Actions $\mathcal{A} = \{1, \ldots, n_a\}$, $\quad A : \mathcal{X} \Rightarrow \mathcal{A}$

Reward function $R : \mathcal{X} \times \mathcal{A} \to \mathbb{R}$

Black-box (probabilistic) transition function $T : \mathcal{X} \times \mathcal{A} \to \mathcal{X}$

Learning rate $\alpha \in [0, 1]$, typically $\alpha = 0.1$

Discounting factor $\gamma \in [0, 1]$

procedure QLEARNING($\mathcal{X}$, A, R, T, α, γ)

Initialize $Q : \mathcal{X} \times \mathcal{A} \to \mathbb{R}$ arbitrarily

while Q is not converged **do**

Start in state $s \in \mathcal{X}$

while s is not terminal **do**

Calculate π according to Q and exploration strategy (e.g. $\pi(x) \leftarrow \arg\max_a Q(x, a)$)

$a \leftarrow \pi(s)$

$r \leftarrow R(s, a)$ ▷ Receive the reward

$s' \leftarrow T(s, a)$ ▷ Receive the new state

$Q(s', a) \leftarrow (1 - \alpha) \cdot Q(s, a) + \alpha \cdot (r + \gamma \cdot \max_{a'} Q(s', a'))$

$s \leftarrow s'$

return Q

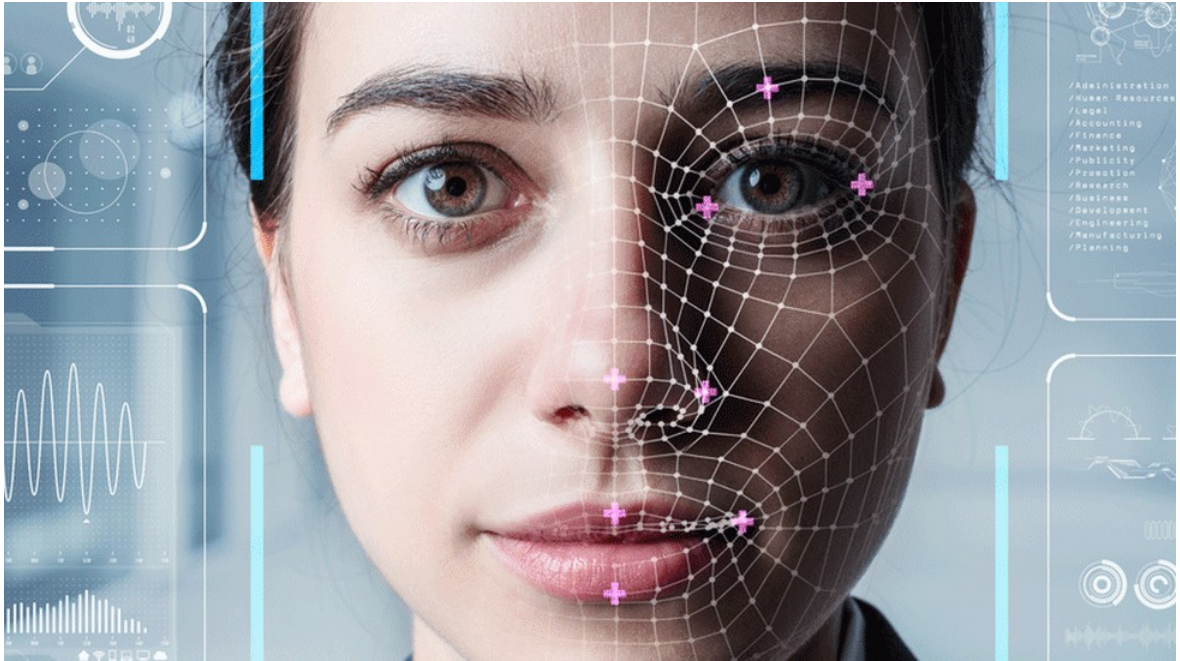
program

C++

Java™

implementation

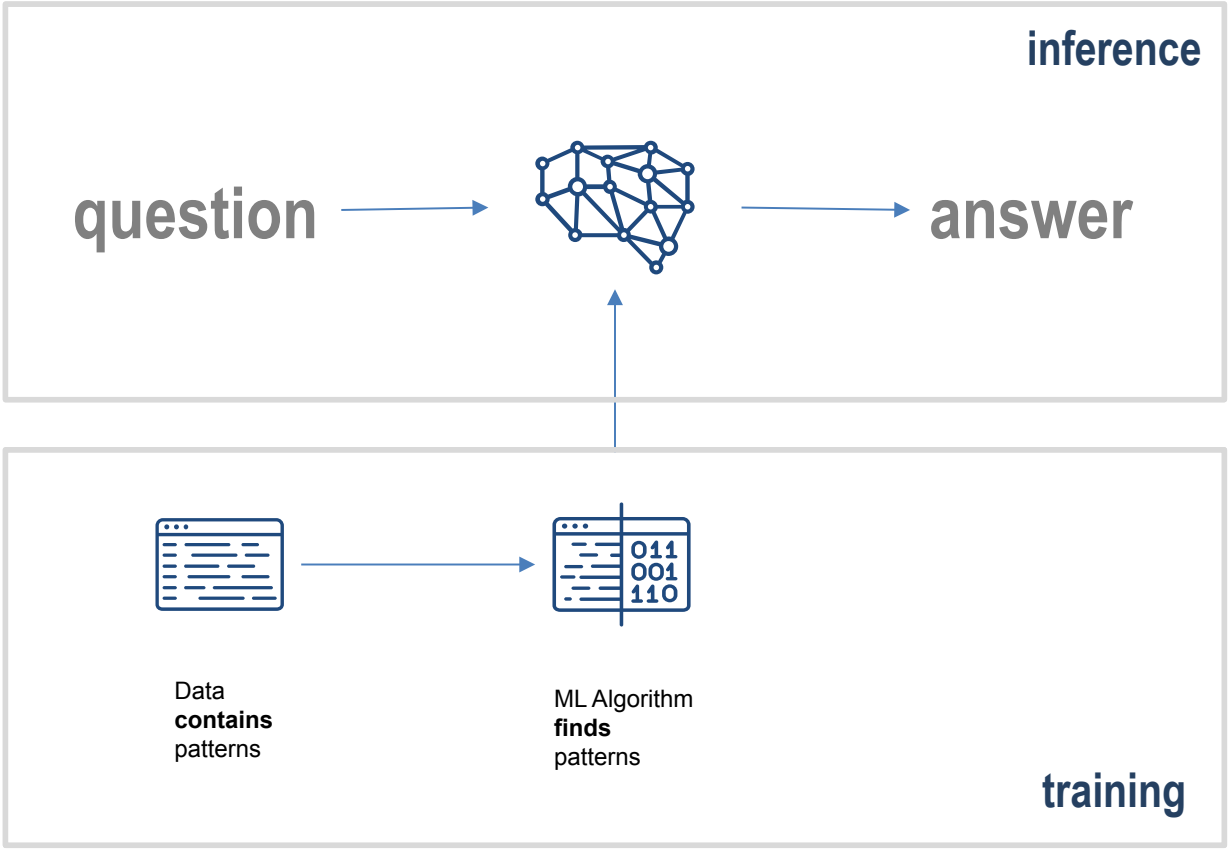
But sometimes algorithm itself is really non-trivial…



Ask machine to create it!

question → → answer

model


inference
question
answer
011
001
110
Data
contains
patterns
ML Algorithm
finds
patterns
training

ML model

student

ML algorithm

teaching methodology

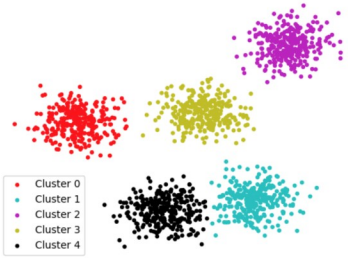
Data Scientist

teacher

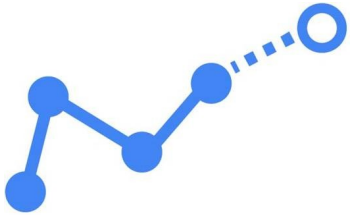
1. The models specialize in solving various problems

classification

segmentation

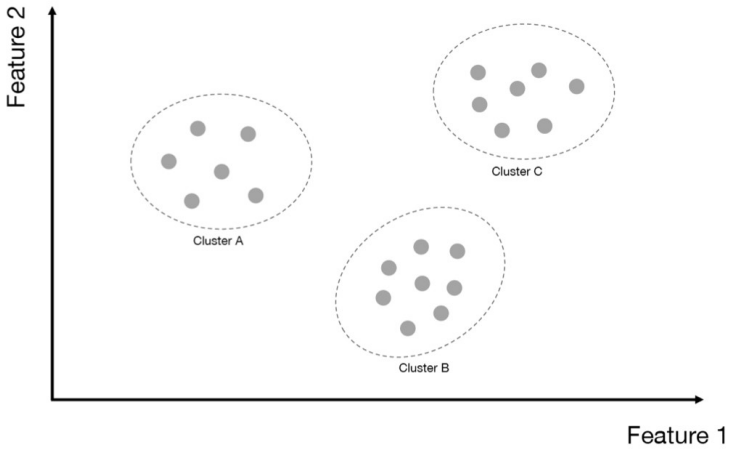


forecasting



Pattern recognition > segmentation

Do *our customers form any groups?*
What *products should we offer them?*
How *do we organize our sales (or support) teams?*



Classification

For each object in the population indicate the categories to which it belongs

Which *of my clients will take advantage of my offer?*

Is it *a cancer, or not?*

Will *this tire break in next 1000 miles?*

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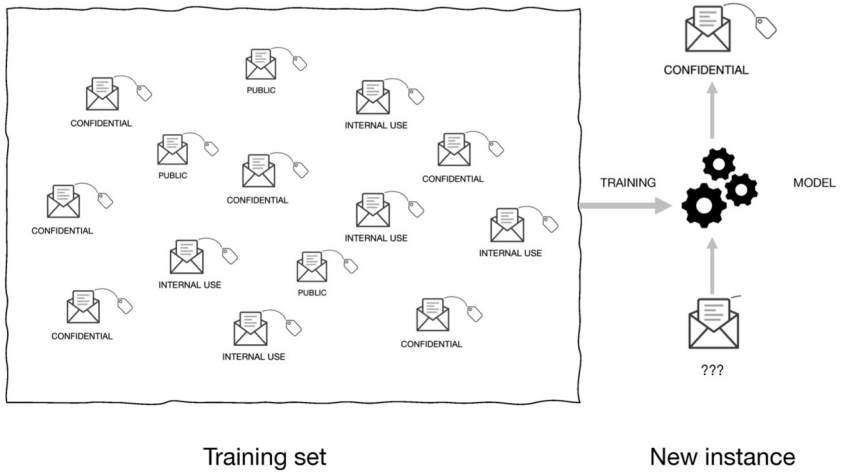
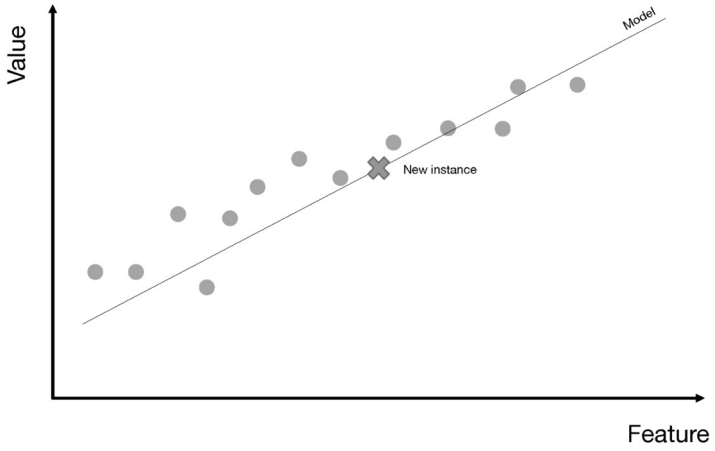


Forecasting > regression

Estimate the numerical value of the selected variable

How many *times will this customer use my offer?*



2. models can be trained in different ways

At the beginning the model is not "smart"... We have to teach it in some way...

beginner

Has no knowledge or experience.
Randomly answers questions

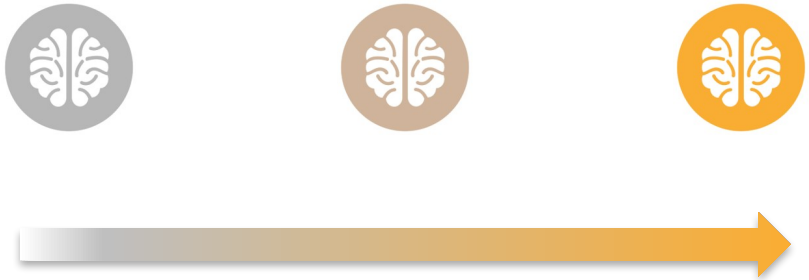
good

Relatively well answers the questions.
However, doesn’t have experience in new situations yet.

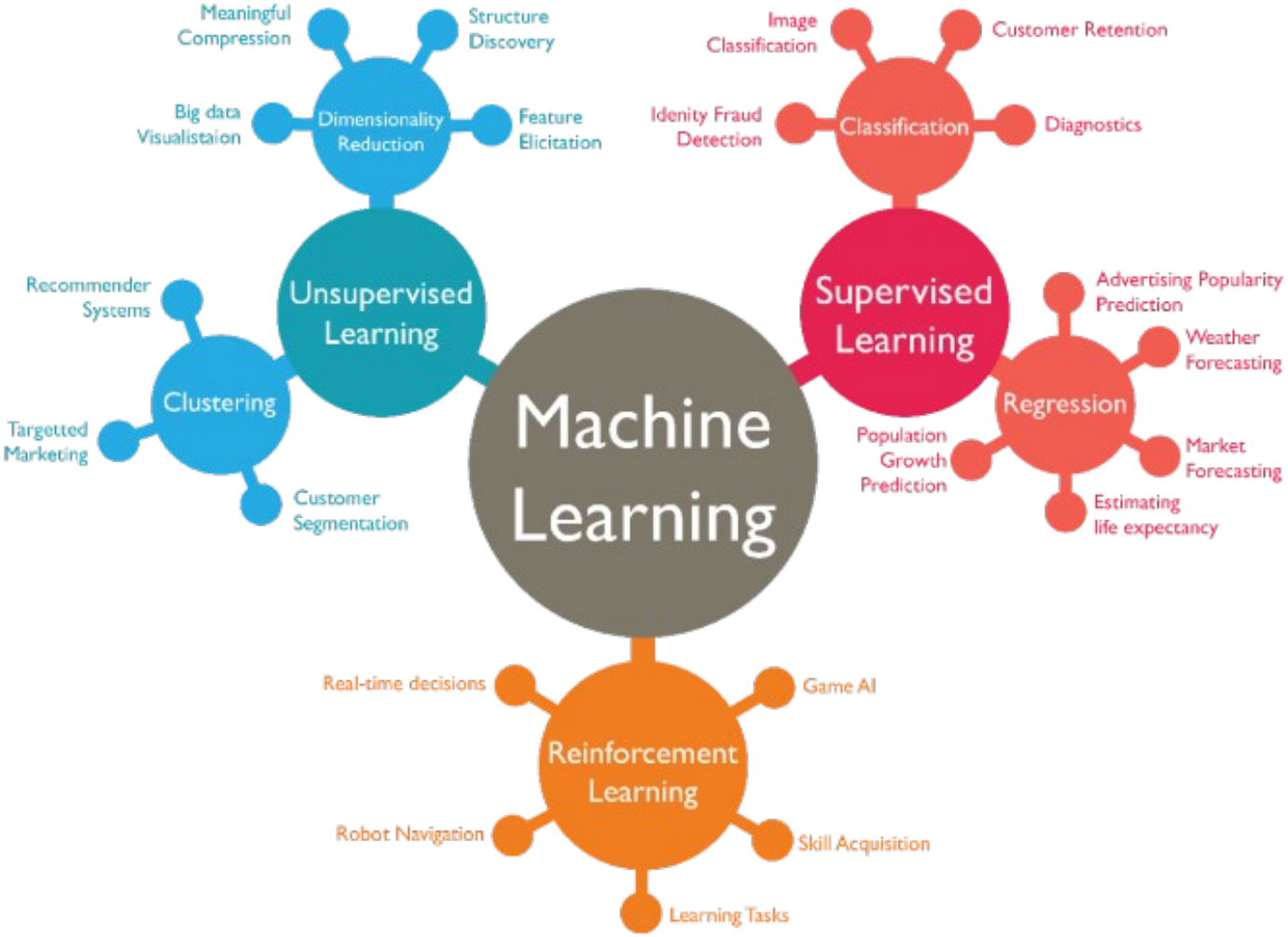
expert

He answers questions perfectly.
Tested in battlefield, learns from new experiences.

Machine Learning Algorithm



set of instructions to the machine, how to train the model


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

Supervised learning

Input: the training set
questions and answers
data + labels

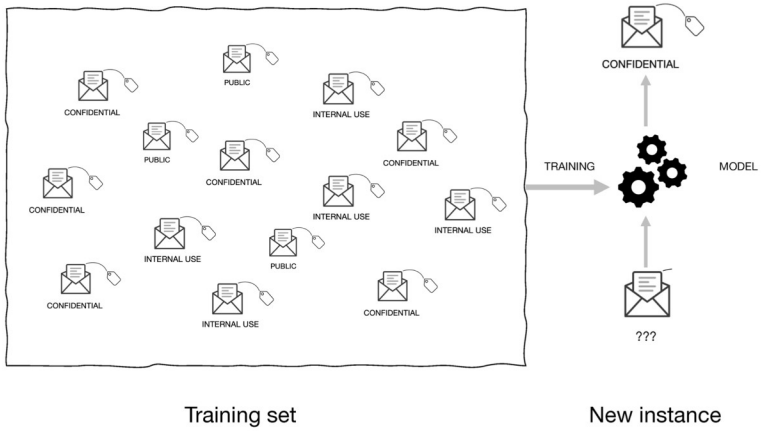
Features and corresponding label presented by the teacher

This is a car
This is a car, because it has four wheels
If it has 2 wheels, we call it a bicycle

Supervised learning

Input: the training set
questions and answers
data + labels

Classification
Regression



Unsupervised learning

Without a teaching set (set of known questions and answers)

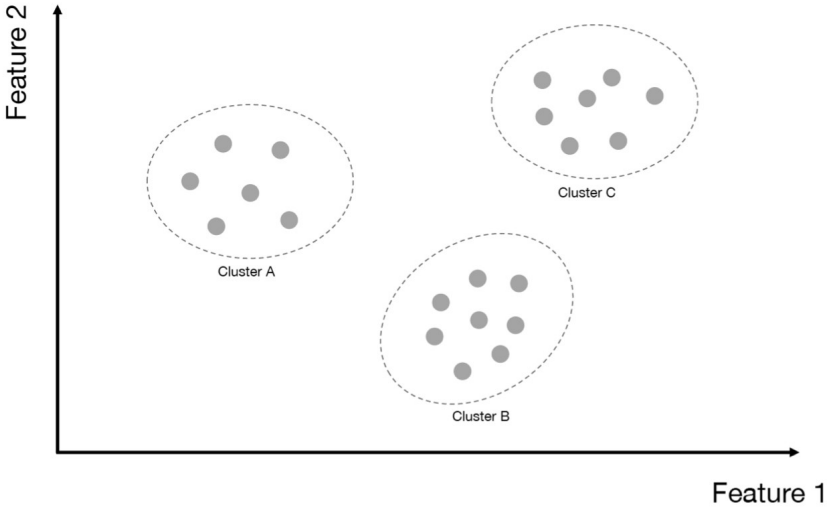
Learner identifies features and patterns by himself…

… group similar objects into clusters

… and assigns his own labels to these clusters

Unsupervised learning

Without a teaching set (set of known questions and answers)



Reinforcement learning

Agent <> Environment
Actions <> Reactions
Goal Function
Reward <> Punishment

Trials & Errors > Learning

Reinforcement learning

Agent <> Environment
Actions <> Reactions
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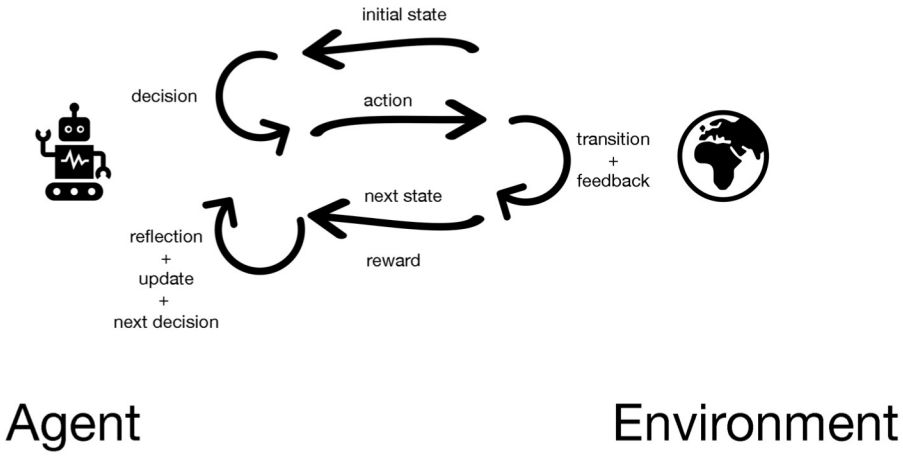
Trials & Errors > Learning



3. models within each of these specialties will be evaluated differently

Regression	Classification	Recommender System
• Mean Absolute Error (MAE) • Root Mean Squared Error (RMSE) • R-Squared and Adjusted R-Squared	• Recall • Precision • F1-Score • Accuracy • Area Under the Curve (AUC)	• Mean Reciprocal Rank • Root Mean Squared Error (RMSE)

Analogy 1

Models?
Algorithms?
Evaluation?

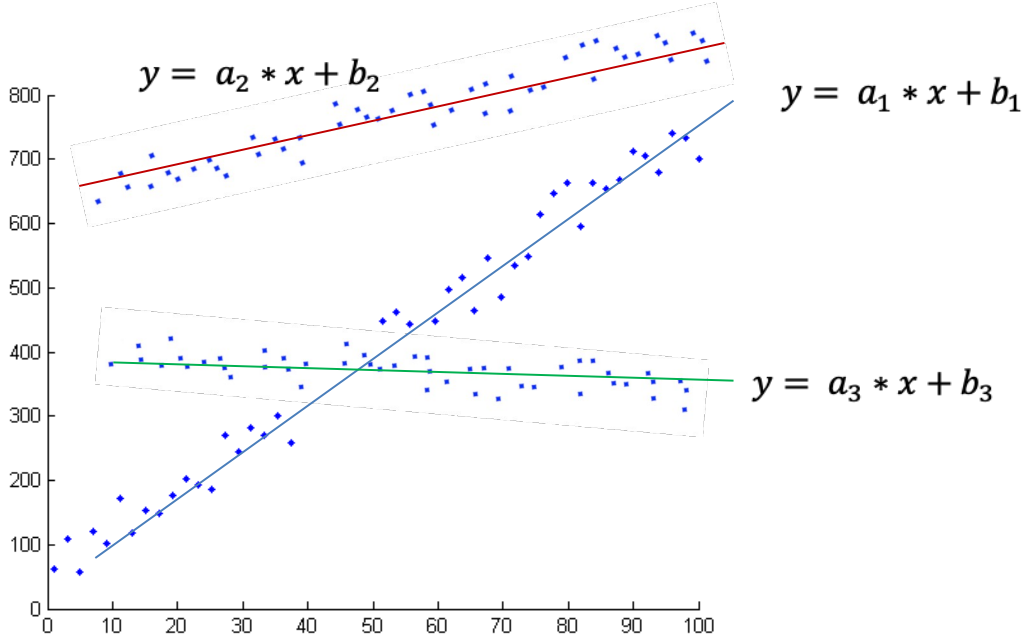
Analogy 2

Models?
Algorithms?
Evaluation?

Analogy 3

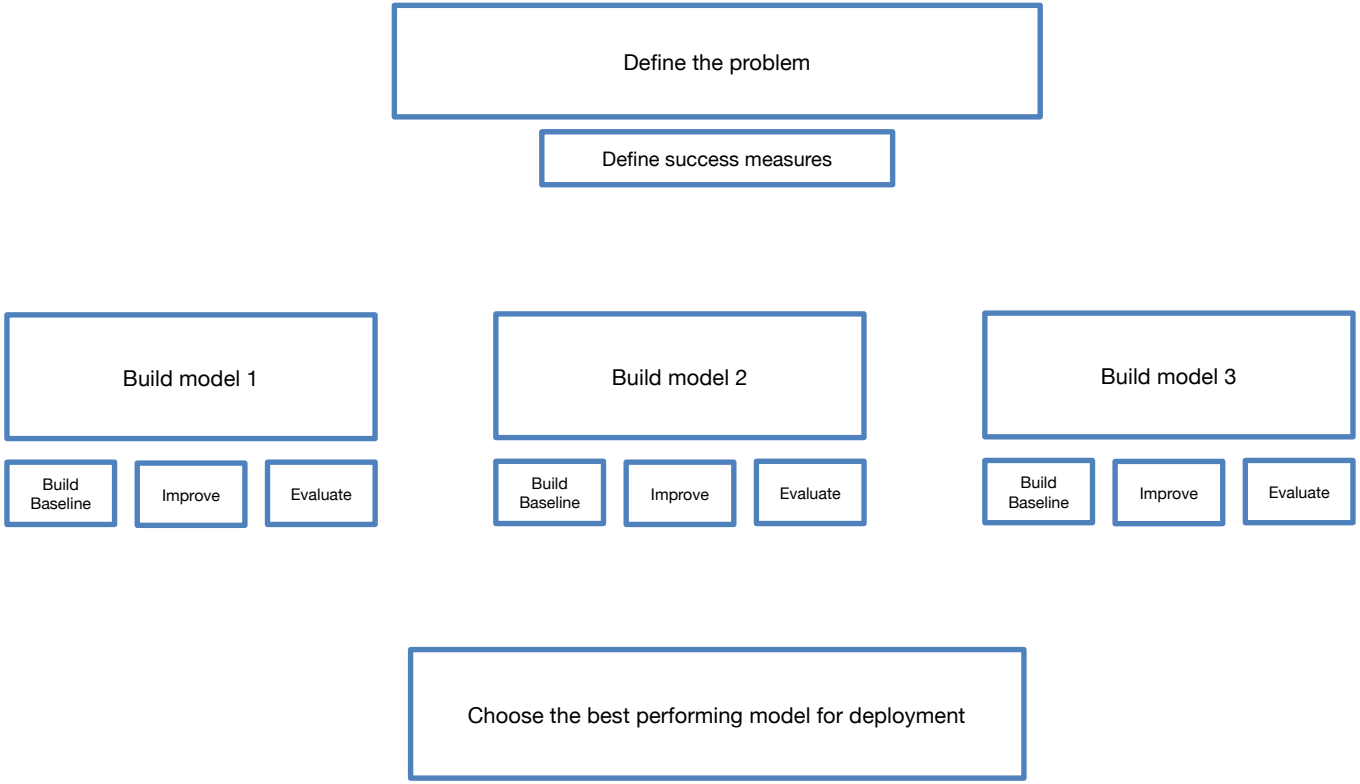
Models?
Algorithms?
Evaluation?

Algorithms vs Models?



Different models. Same algorithm.

How does it work in practice?


Define the problem
Define success measures
Build model 1
Build Baseline
Improve
Evaluate
Build model 2
Build Baseline
Improve
Evaluate
Build model 3
Build Baseline
Improve
Evaluate
Choose the best performing model for deployment